# Solution Requirements

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| **Date** | 26 June 2026 |
| **Team ID** | |
| **Project Name** | A Comprehensive Measure of Well-Being |
| **Maximum Marks** | 4 Marks |

**Functional and Non-Functional Requirements:**

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

**Step 1: Functional Requirements (FR)**

| S.No | Requirement Category | Requirement Description | Priority (High/Medium/Low) |
|---|---|---|---|
| 1 | Authentication | *The application allows users to access the HDI prediction system through the web interface without complex authentication.* | Medium |
| 2 | Authorization levels | *All users have permission to enter input data and generate HDI predictions.* | Low |
| 3 | External interfaces | *The system interacts with the Flask web application, HTML pages, and the trained HDI.pkl machine learning model.* | High |

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| **4** | Transactions processing | *The system accepts user inputs, processes them using the machine learning model, and generates an HDI prediction instantly.* | High |
| **5** | Reporting | *The system displays the predicted HDI score along with its development category (Low, Medium, High, or Very High).* | High |
| **6** | Business rules, etc. | *Prediction is based on Country, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita. Invalid inputs are rejected.* | High |
| **7** | Compliance to laws or regulations | *The application follows basic software development practices and ensures that no personal user information is permanently stored.* | Medium |
| **8** | *Other* | *The system provides a simple, responsive interface that enables users to perform predictions quickly and accurately.* | Medium |

### Step 2: Non-Functional Requirements (NFR)

| **S.No** | **NFR Category** | **Requirement Description** | **Target Metric / Acceptance Criteria** |
|---|---|---|---|
| **1** | Performance & Speed | *The system should generate predictions quickly after receiving valid inputs.* | *Prediction time less than 2 seconds* |
| **2** | Scalability | *The application should support multiple users with minimal performance degradation.* | *Supports multiple concurrent users* |
| **3** | Security & Data Privacy | *User inputs are processed securely without permanent storage of personal information.* | *No sensitive data stored permanently* |
| **4** | Reliability & Availability | *The application should provide consistent prediction results and remain available during normal operation.* | *99% application availability* |
| **5** | Usability & Accessibility | *The interface should be simple, responsive, and easy to understand for all users.* | *User-friendly interface with clear navigation* |
| **6** | *Other* | *The application should be maintainable, reusable, and easy to update with improved machine learning models.* | Modular code structure and documentation |